

SELF-DECLARATION

We declare that this report and the associated implementation represent our team's original work for the CS/EE5841 final project. All external research, software libraries, and technical sources used in this work have been cited where appropriate.

Jah Multani: Developed the primary data-processing and residual learning notebook, implemented and evaluated the MLP and 1D CNN models, analyzed model performance, and contributed to the report and presentation.

Jevin Barnett: Contributed to data collection and preparation, classical filtering methods, additional neural-network experiments, results analysis, and preparation of the report and presentation.

Kyle Hildebrand: Contributed to the denoising autoencoder and IIR filter experiments, STM32 deployment and timing analysis, interpretation of embedded results, and preparation of the report and presentation.

All team members reviewed the final implementation, report, repository, and presentation.